JFET THEORY OF OPERATION

INTRODUCTION

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORTORY

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

JFET Construction

Regions of Operation

Transconductance Curve

JFET CONSTRUCTION

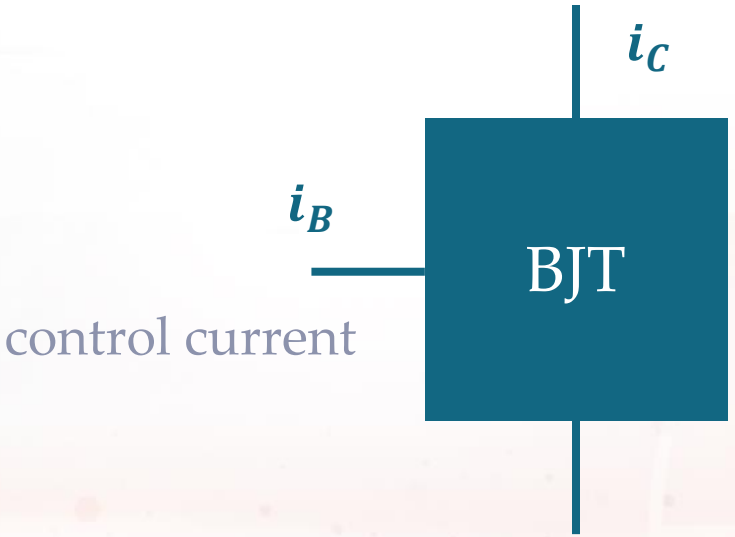
1955

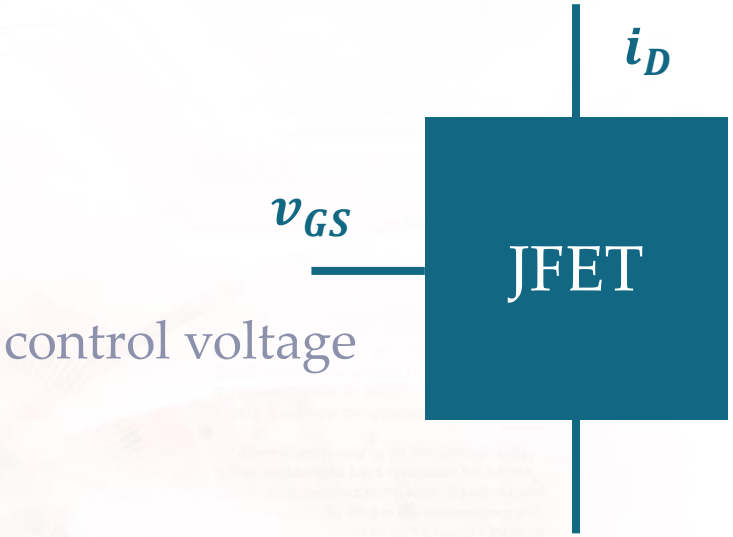
Dr. Ian Munro Ross

Dr. G.C. Dacey

Junction Field Effect Transistor

BJT vs JFET





Current-controlled device

Bipolar device

electron and holes

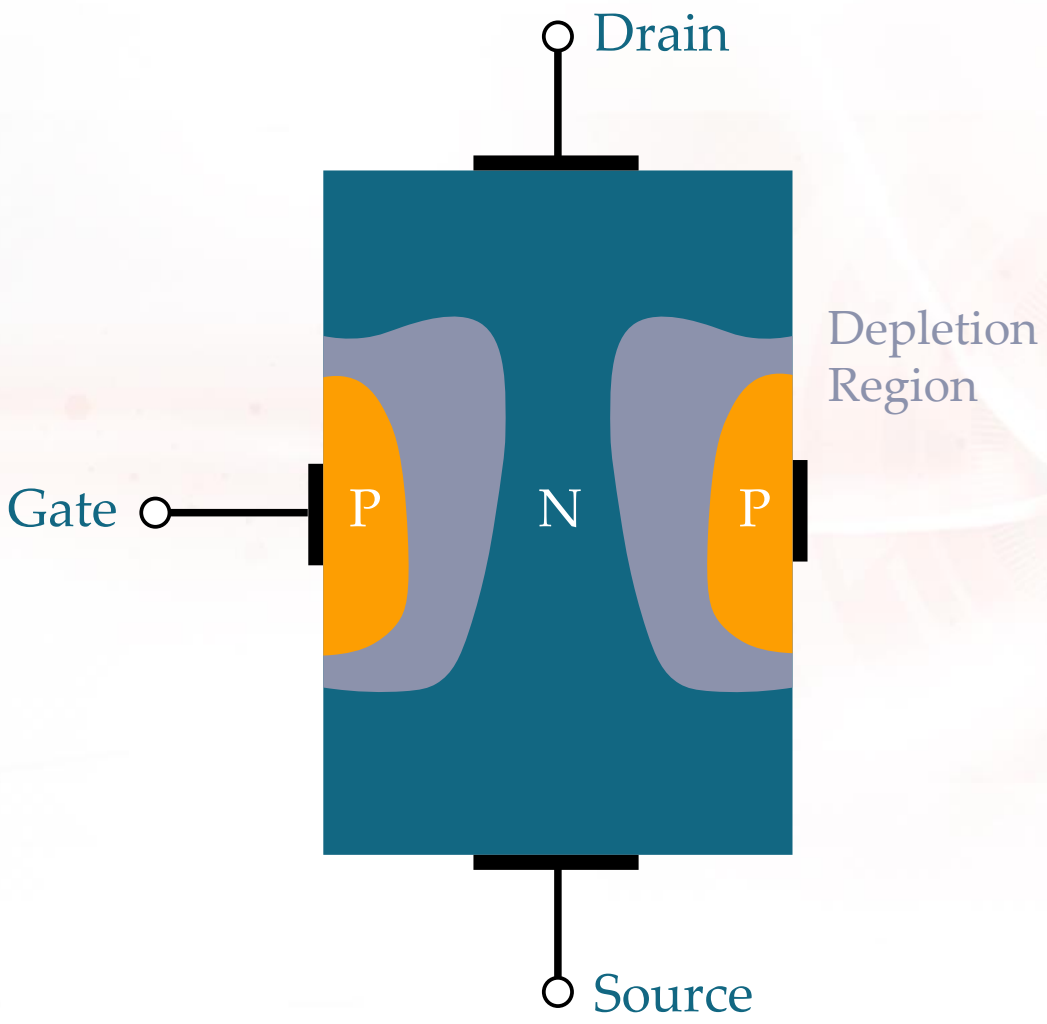
Voltage-controlled device

Unipolar device

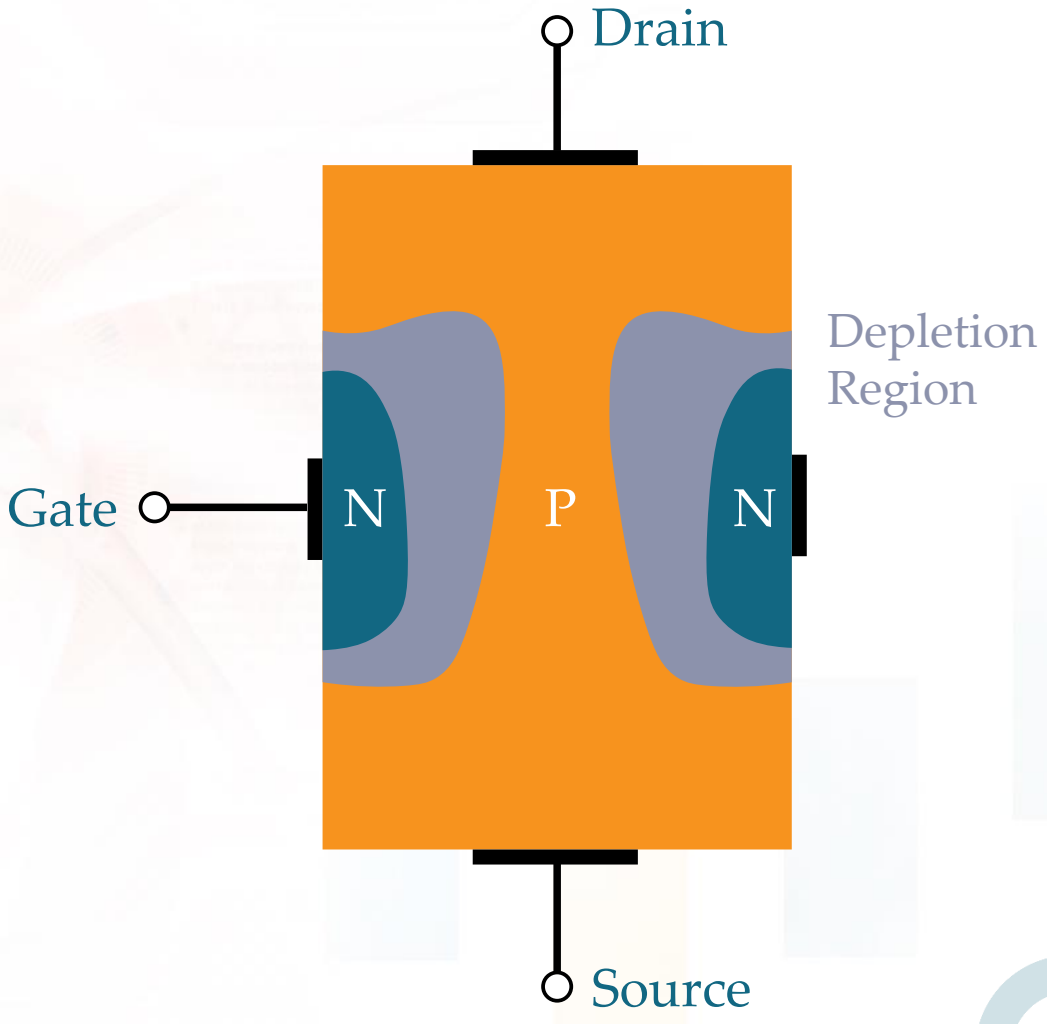
electron

CONSTRUCTION

N-Chanel



P-Channel



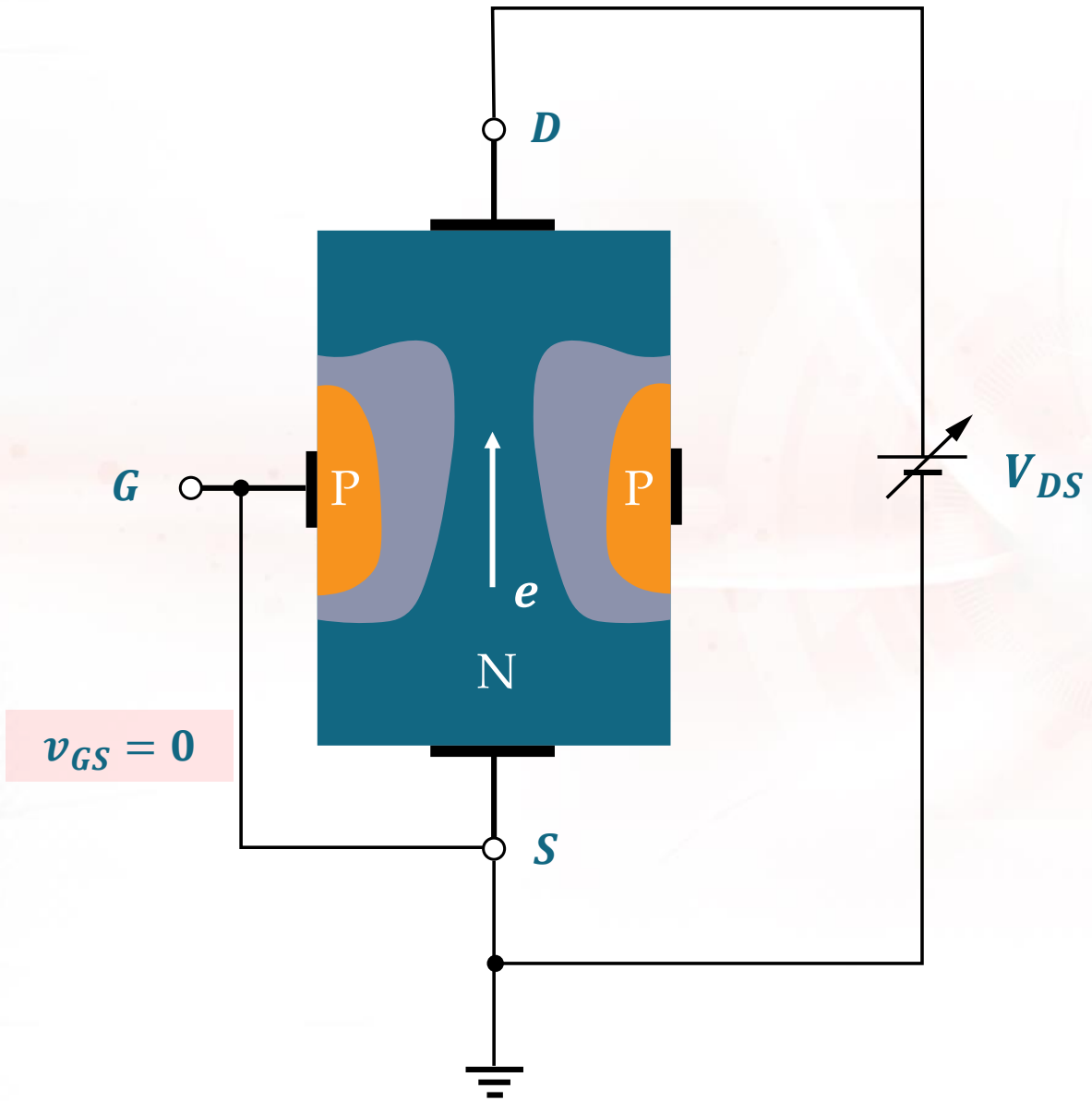
SCHEMATIC SYMBOL

N-Chanel

P-Channel

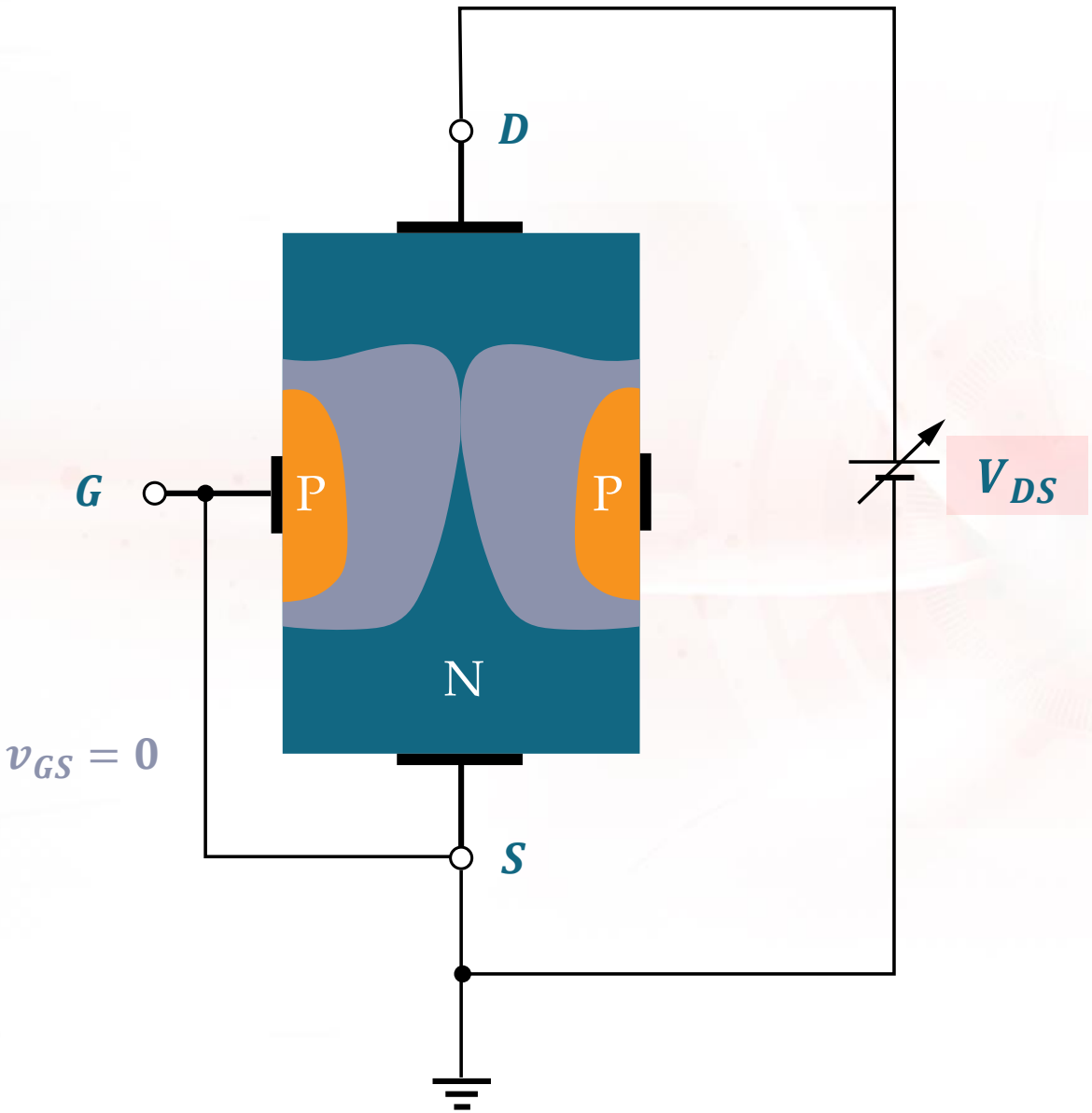
REGIONS OF OPERATION

NORMALLY ON DEVICE



Whe v_{GS} is zero, **maximum drain current** flows between the source and the drain

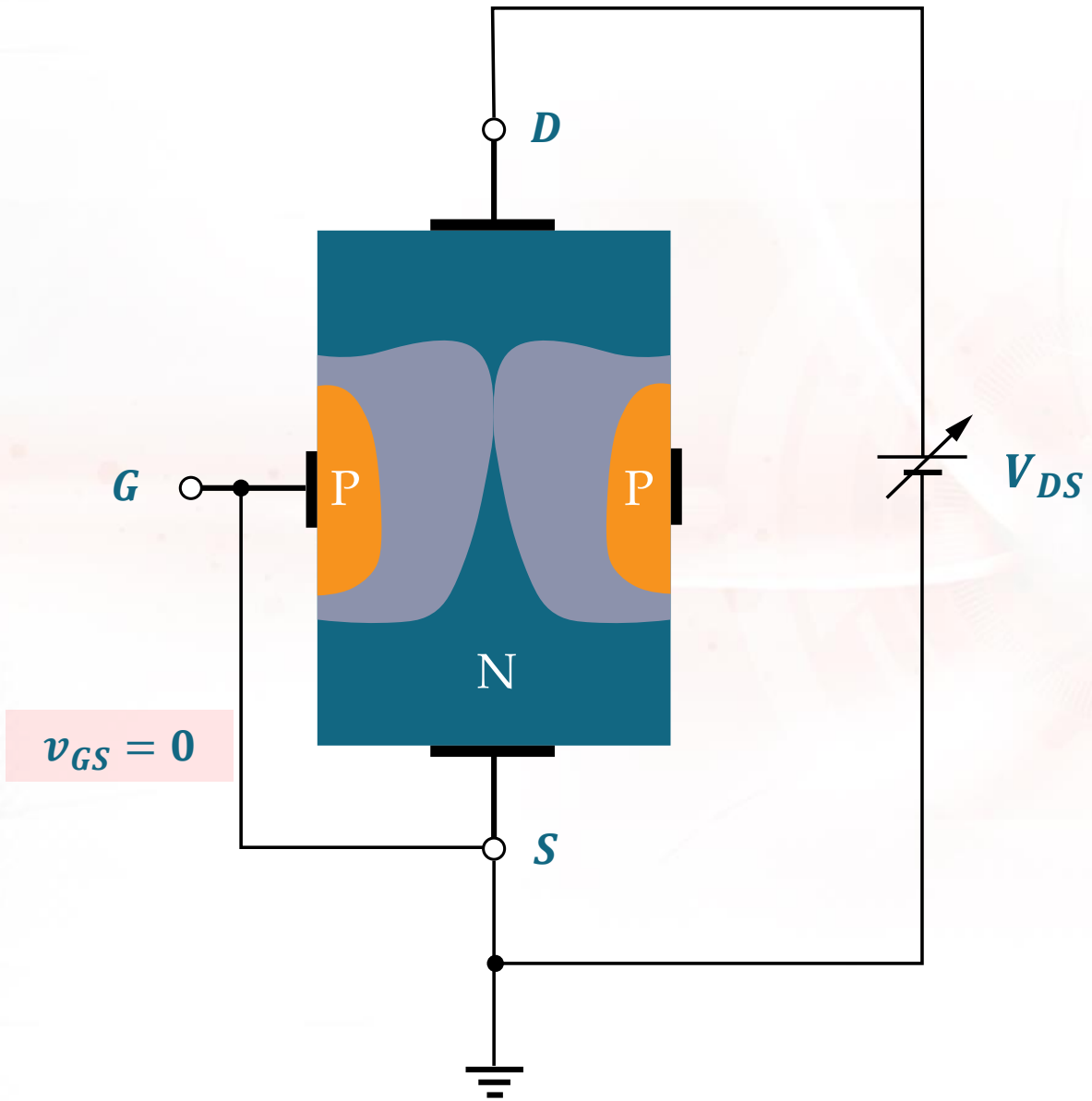
PINCH-OFF VOLTAGE



If V_{DS} is increased to a level where it appears that two regions would "touch", a condition referred to as **pinch-off** will result

The level of V_{DS} that established this condition is referred to as **pinch-off voltage, v_P**

SATURATION LEVEL

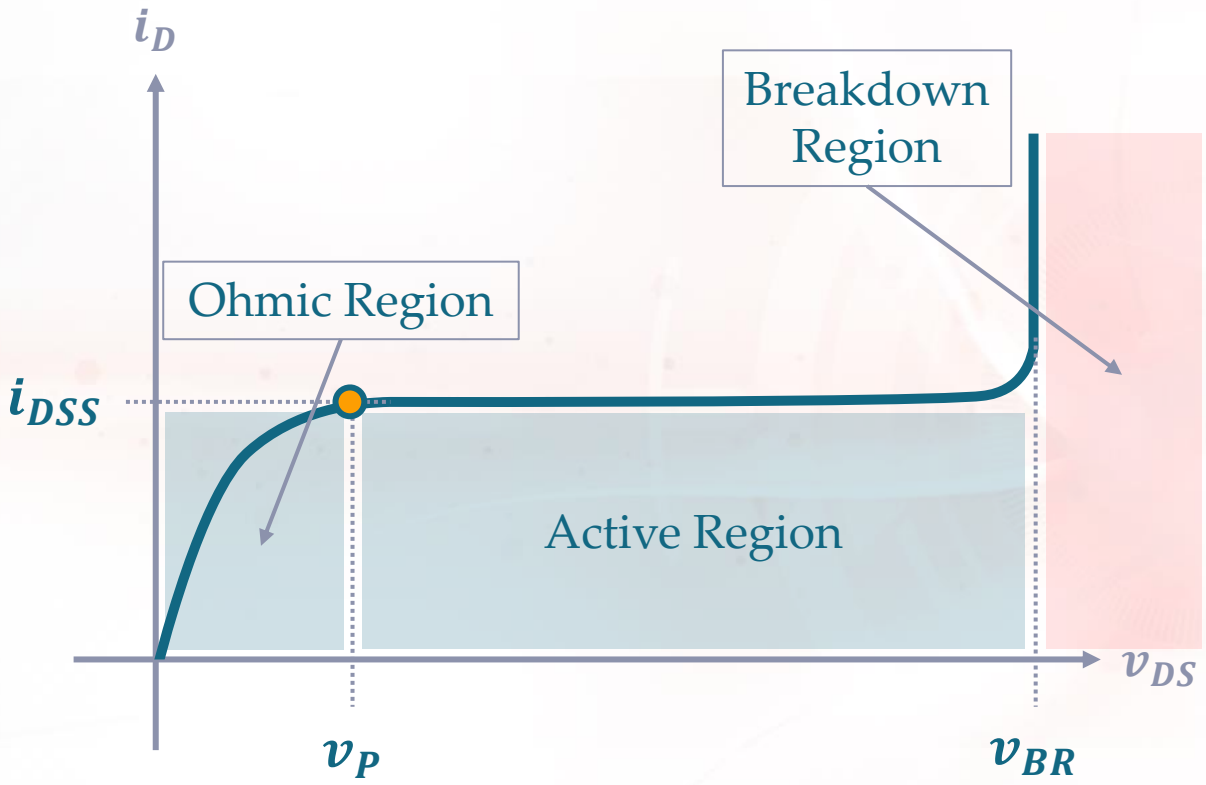


i_D does not drop off at pinch-off and maintains **saturation level** defined as i_{DSS}

Drain-to-Source current with a Short-circuit connection from gate to source

The **maximum drain current**

DRAIN CURVE



When operated in the **ohmic region**, a JFET is equivalent to a resistor with a value of approximately:

$$R_{DS} = \frac{v_P}{i_{DSS}}$$

An MPF4857 has $v_p = 6\ V$ and $i_{DSS} = 100\ mA$.

What is the **ohmic resistance**? The **gate-source cutoff voltage**?

The ohmic resistance

$$R_{DS} = \frac{v_P}{i_{DSS}}$$

$$R_{DS} = \frac{6\ V}{100\ mA}$$

$$\boxed{R_{DS} = 60\ R}$$

ans

The gate-source cutoff voltage

$$v_{GS(off)} = -v_p$$

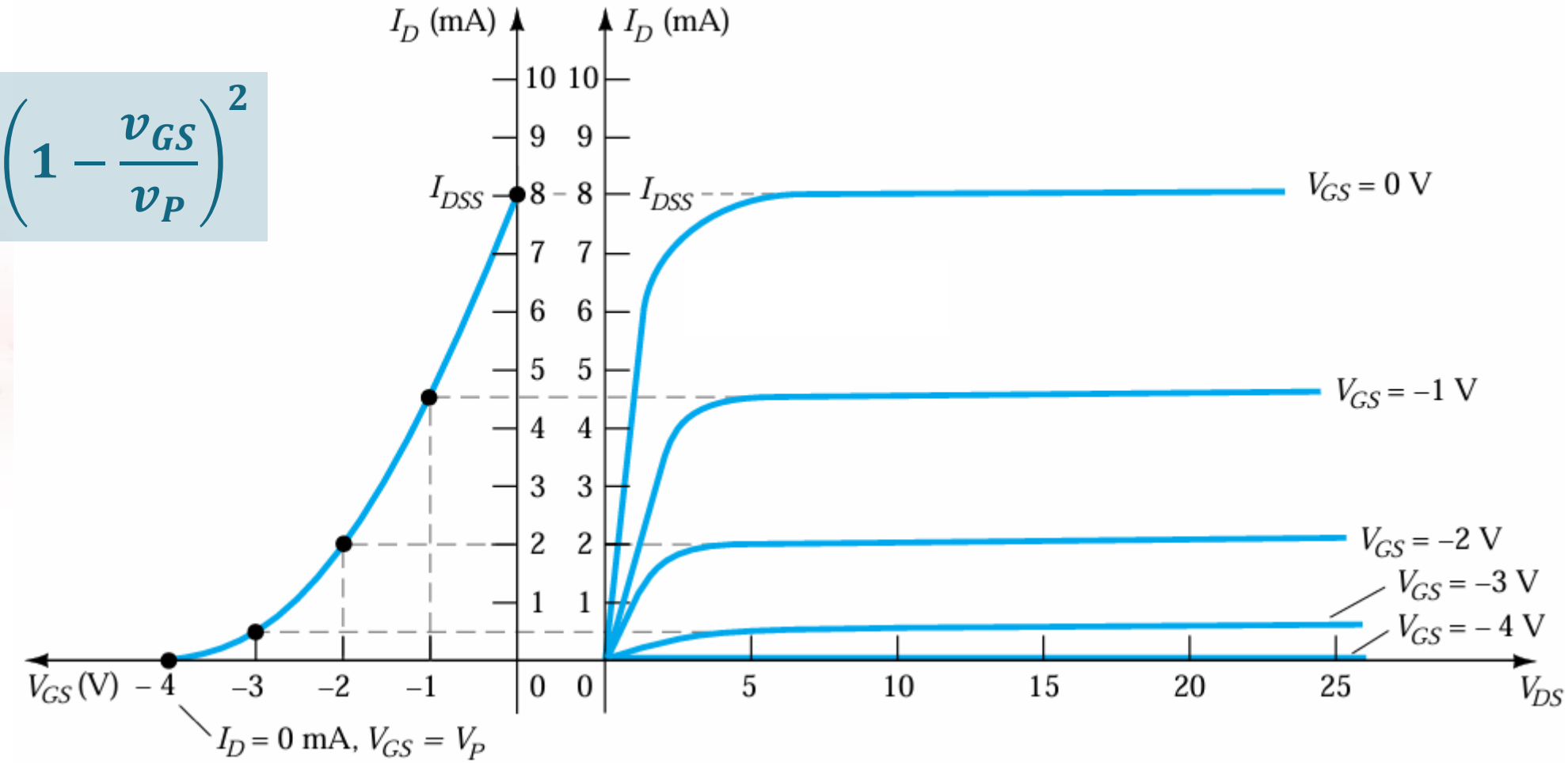
$$\boxed{v_{GS(off)} = -6\ V}$$

ans

TRANSCONDUCTANCE CURVE

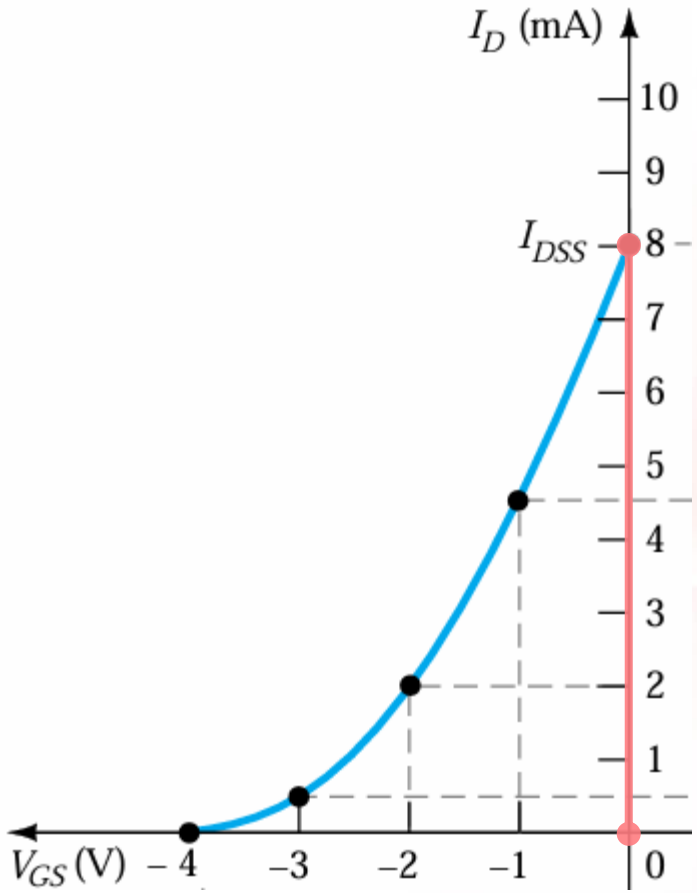
SHOCKLEY'S EQUATION

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$



Transconductance Curve

Drain Curve



Transconductance Curve

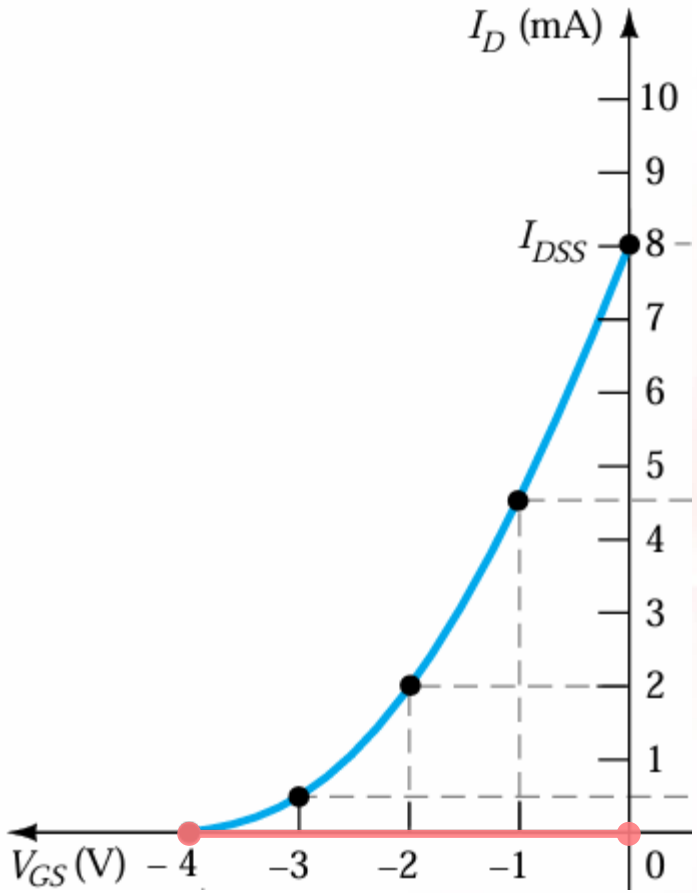
$$i_D = i_{DSS}\Big|_{v_{GS}=0}$$

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = i_{DSS}\left(1 - \frac{0}{v_P}\right)^2$$

$$i_D = i_{DSS}(1 - 0)^2$$

$$i_D = i_{DSS}$$



Transconductance Curve

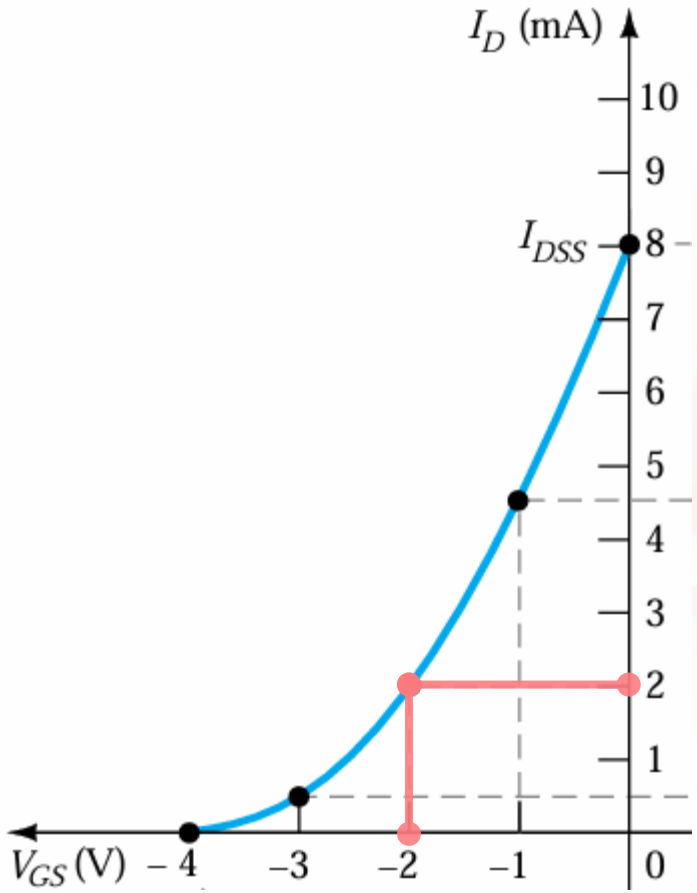
$$i_D = 0\Big|_{v_{GS}=v_P}$$

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_{GS}}\right)^2$$

$$i_D = i_{DSS}(1-1)^2$$

$$i_D = 0$$



Transconductance Curve

$$i_D = \frac{1}{4} i_{DSS}\Big|_{v_{GS}=0.5v_P}$$

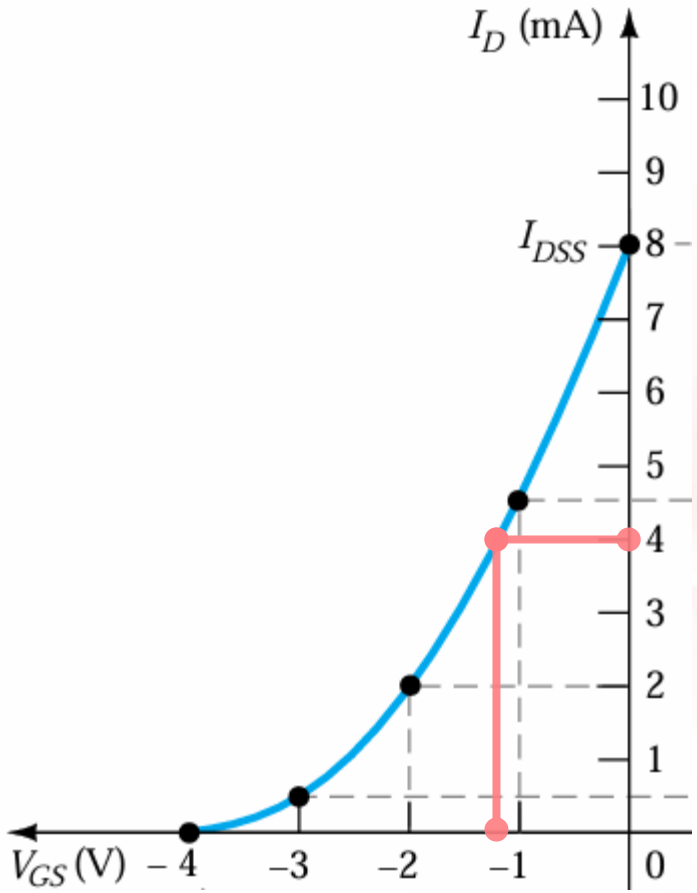
$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = i_{DSS}\left(1 - \frac{0.5v_P}{v_p}\right)^2$$

$$i_D = i_{DSS}(1 - 0.5)^2$$

$$i_D = i_{DSS}(0.5)^2$$

$$i_D = 0.25 i_{DSS}$$



Transconductance Curve

$$v_{GS} = 0.3 v_P \Big|_{i_D = 0.5 i_{DSS}}$$

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$0.5 i_{DSS} = i_{DSS}\left(1 - \frac{v_{GS}}{v_p}\right)^2$$

$$\sqrt{0.5} = 1 - \frac{v_{GS}}{v_P}$$

$$\frac{v_{GS}}{v_P} = 1 - \sqrt{0.5}$$

$$v_{GS} \approx 0.3 i_{DSS}$$

Sketch the transconductance curve defined by

$\boldsymbol{I_{DSS} = 12mA}$ and $\boldsymbol{V_P = -6V}$.

i_D @ $v_{GS} = 0.5v_P$

$$\boldsymbol{i_D = \frac{1}{4} i_{DSS}}\Big|_{v_{GS}=-3}$$

$$\boldsymbol{i_D = \frac{1}{4}(12\ mA)}\Big|_{v_{GS}=-3}$$

$$\boxed{\boldsymbol{i_D = 3\ mA}\Big|_{v_{GS}=-3}}$$

ans

v_{GS} @ $i_D = 0.5i_{DSS}$

$$\boldsymbol{v_{GS} = 0.3v_P}\Big|_{i_{DSS}=6\ mA}$$

$$\boldsymbol{v_{GS} = 0.3(-6)}\Big|_{i_{DSS}=6\ mA}$$

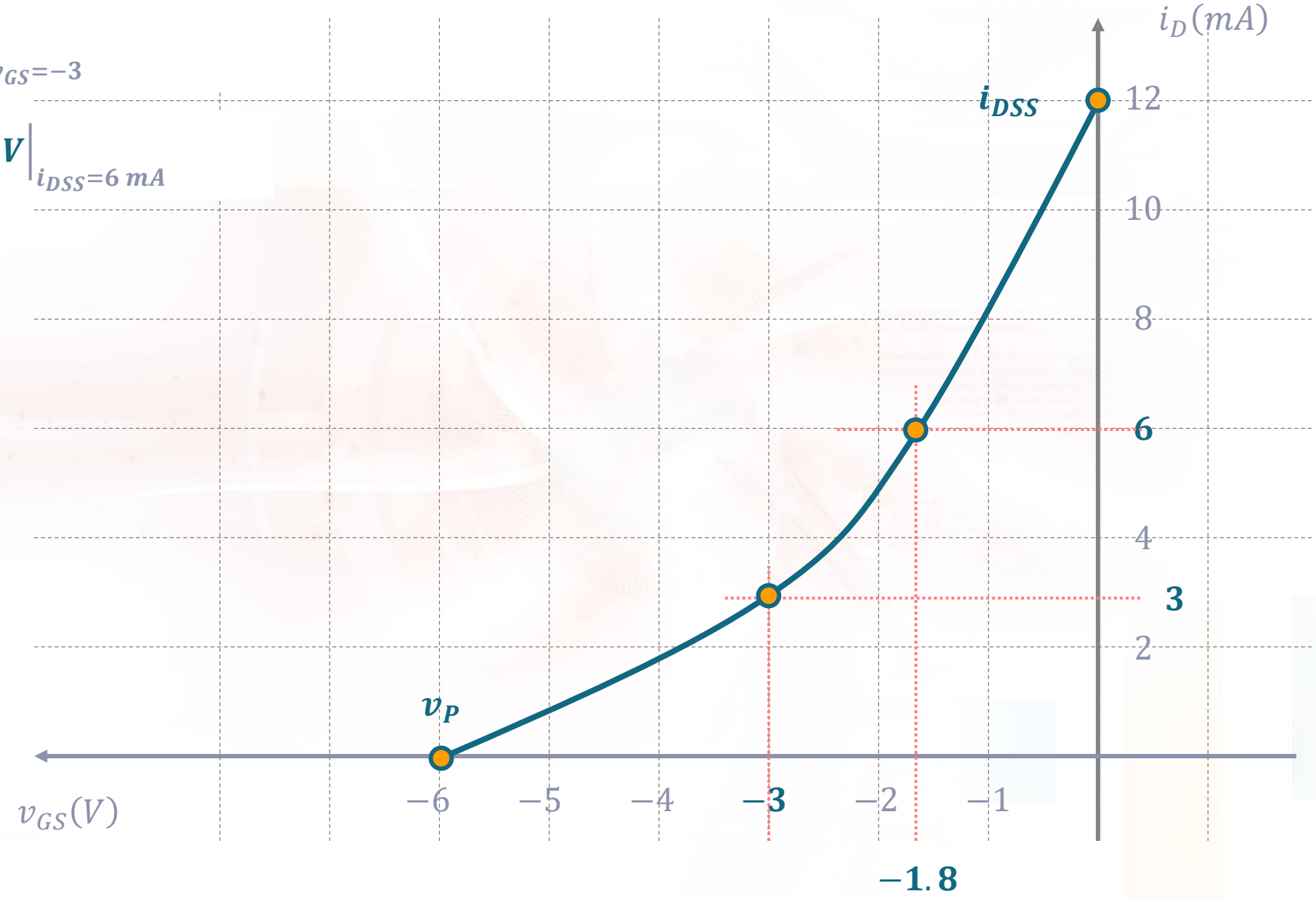
$$\boxed{\boldsymbol{v_{GS} = -1.8\ V}\Big|_{i_{DSS}=6\ mA}}$$

ans

TRANSCONDUCTANCE CURVE

$$i_D = 3\ mA\Big|_{v_{GS}=-3}$$

$$v_{GS} = -1.8\ V\Big|_{i_{DSS}=6\ mA}$$



LABORATORY